

P SANGAM

AI Software Engineer

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SUMMARY

AI Software Engineer and individual contributor who builds end-to-end AI systems as proofs of concept and pilots, spanning time-series forecasting, real-time computer vision, IoT-to-cloud pipelines, and LLM-based agentic workflows. At EAGE Technologies, worked largely independently in a small team, taking projects from requirements to working prototypes: forecasting models that scored 89% and 80% on evaluation, a computer vision pipeline integrated with hardware, and agentic pipelines built with LangChain, LangGraph, and MCP. Recently built EV Ops Copilot end to end: an agentic RAG assistant with SQL governance, hybrid search, grounded answers, monitoring, and cost controls. Comfortable explaining technical trade-offs to non-technical stakeholders through clear technical demos. Strong foundations in classical ML, deep learning, and Generative AI, now focused on production-grade LLM systems.

CORE SKILLS

AI & ML: Machine Learning, Deep Learning, Computer Vision, CNNs, Transformers, Time-Series Forecasting, Statistics & Data Science, LightGBM, LSTM, OpenCV, YOLOv8

LLM & Agentic Engineering: Agentic Pipelines (LangChain, LangGraph), MCP, Tool Calling, Function Calling, RAG, Hybrid Search & Re-ranking, Prompt Engineering, LLM Guardrails, Vector DBs (FAISS, pgvector)

Evaluation & Observability: LLM-as-Judge, Golden Datasets, Regression Testing, Groundedness & Hallucination Checks

Tools & Environments: Cursor, Claude Code, Antigravity, Hermes, Docker, FastAPI, Git, Langfuse

Delivery & Collaboration: Technical Demos, Requirements Gathering, Agile/Scrum, Stakeholder Communication

Software & Systems: Python, ReactJS, SQL, PostgreSQL

EXPERIENCE

EAGE Technologies India Pvt. Ltd.

Bengaluru, KA

AI Software Engineer

Jul 2025 – Present

- Built and benchmarked LightGBM and LSTM forecasting models for solar power output using 12 years of location-based historical data, reaching evaluation scores of 89% and 80% respectively, and designed retraining triggers and performance-ratio-based drift monitoring.
- Architected an agentic decision pipeline integrating LightGBM/LSTM forecasts with rule-based risk logic and LLM function calling, generating real-time operational directives and automated risk alerts for solar assets.
- Built a real-time computer vision pipeline (YOLOv8, InsightFace, FAISS) for face detection, distance estimation, and object detection, engineered as an independent software pipeline, with STM32 hardware connected to the system to complete the end-to-end hardware-software pipeline for a client defense application.
- Designed an IoT-to-cloud Agri-Soil Intelligence engine under zero-historical-data constraints, engineering cold-start threshold logic and location-based climate API pipelines for 2 pilot locations.
- Interfaced directly with client stakeholders and engineering leads to translate complex operational needs into technical specifications and deploying pilot systems in live customer environments.
- Conducted applied research and development of machine learning and deep learning solutions across predictive analytics, computer vision, and forecasting use cases.

Code Nimbus Solutions Pvt. Ltd.

Bengaluru, KA

AI & Frontend Intern

Sep 2024 – Jun 2025

- Built AI-powered facial recognition pipelines using deep learning and embedding extraction techniques.
- Developed user interfaces for AI applications, enabling visualization of model outputs and operational workflows.

PROJECTS

EV Ops Copilot: End-to-end agentic RAG assistant for EV ops teams that diagnoses vehicle issues and answers sales questions from telemetry, service records, and documents. Built with LangGraph and MCP: hybrid search with re-ranking, grounded cited answers, SQL guardrails (AST validation, cost gating, read-only PostgreSQL/pgvector), prompt-injection-safe ingestion, cost-aware model tiering with fallbacks, Langfuse monitoring, Docker/CI deployment, and golden-dataset regression evals.

Solar Intelligence Platform: Forecasting and decision-support platform combining LightGBM and LSTM models (89% / 80% evaluation scores) trained on 12 years of historical data with rule-based risk analysis and an LLM-generated advisory layer, designed with retraining triggers and drift-detection monitoring.

Soil Intelligence Platform: IoT-to-cloud agricultural advisory system using GSM-based sensor data and location/climate APIs, threshold-based rule engine for soil health, crop recommendation, and disease risk across two pilot locations, built in the absence of historical data.

Defense Surveillance: Real-time computer vision system integrating pretrained YOLOv8, InsightFace, and FAISS for face detection, distance estimation, and object detection (including IR images), built as a standalone software pipeline, with STM32 embedded hardware connected to the system to complete the hardware-software pipeline.

EDUCATION

SDM Institute of Technology

Ujire, Dakshina Kannada, KA, India

B.E. in Artificial Intelligence & Data Science · 2021 – 2025 · CGPA: 7.69